

Predicting Hospital Readmissions for Diabetic Patients

Technical Report

General Assembly Data Analytics Bootcamp (DABPT03)

Ram Jeevan | February 2026

Executive Summary

This report documents the development of machine learning models to predict hospital readmissions for diabetic patients. Two prediction targets were modelled from a UCI dataset containing records of 101,766 diabetes patient admission encounters:

- Primary Model: Any readmission — LightGBM achieves AUC 0.692, F1 0.595, Recall 55.9%
- Secondary Model: Readmission within 30 days — LightGBM with class balancing and threshold tuning achieves AUC 0.67, F1 0.27, and Recall 49.7%. The 30-day model required special handling (tuning the threshold for readmission to 0.3 rather than 0.5) due to severe class imbalance (only 11.4% positive class).

The models assign each patient a risk score (0-100%) at discharge, enabling clinical teams to structure care packages for individuals based on their probability to readmit and hospital planning teams to forecast the resources they will need for the coming month.

Table of Contents

1. Introduction
2. Data Source and Data Dictionary
3. Data Cleaning
4. Feature Engineering
5. Exploratory Data Analysis
6. Modelling Approach
7. Results: Any Readmission Model
8. Results: 30-Day Readmission Model
9. Interactive Dashboard Application
10. Conclusions and Recommendations
11. Limitations

1. Introduction

1.1 Problem Statement

Hospital readmissions represent a significant burden on healthcare systems. In Singapore, public hospitals operate at 85-95% bed occupancy, meaning that admissions could pose significant problems to the healthcare system. Diabetic patients have elevated readmission rates due to the complexity of managing their condition.

1.2 Objective

Develop 2 risk scoring systems that predict readmission likelihood at discharge:

- Any readmission: Identify patients likely to return at any point.
- Early readmission (<30 days): Identify patients at risk of rapid return, indicating potential care quality issues.

1.3 Success Metrics

To evaluate the performance of our risk-scoring systems, I focused on four primary metrics. These measures ensure the models provide clinical value rather than just mathematical accuracy.

- **AUC-ROC:** Measures the model's ability to distinguish between patients who will be readmitted and those who will not. I aimed for a score close to 0.7 to ensure the model performs significantly better than a random guess.
 - **Recall:** The percentage of actual readmissions the model correctly identifies. In a clinical setting, catching more at-risk patients is vital, so I target a recall of **55%** for the primary model and **50%** for the 30-day model.
 - **F1-Score:** The harmonic mean of precision and recall. This balances our need to find at-risk patients without flagging every single person as a risk.
 - **Precision:** The accuracy of our "positive" flags. While we prioritise recall, we aim to maintain precision high enough to avoid "alert fatigue" among clinical staff.
-

1.4 Approach

I selected three distinct algorithms capturing both simple linear relationships and complex, non-linear patterns in the patient data.

- **Logistic Regression:** Logistic Regression takes all factors and assigns them coefficient weights, then gives a result between 0 and 1. This model allows clinicians to see exactly how specific factors, such as a patient's age or number of previous inpatient visits, directly increase or decrease the risk score. I used 1000 iterations rather than the standard 100 because the dataset is large, I need more iterations to actually get to the

result.

- **Random Forest:** This model uses 100 decision trees to make predictions. It is robust against outliers and handles the categorical nature of medical data well, such as various medication dosage changes or different hospital specialties. Random Forest is good at handling outliers and extreme values and flagging them out as high risk. I used 100 estimators as that gives a reliable percentage as the result. Going beyond that rarely changes the result meaningfully.
- **LightGBM (Light Gradient Boosting Machine):** This is the primary model due to its superior speed and accuracy. It is particularly effective for this dataset because:
 - **Native Categorical Support:** It handles variables like "Race" or "Primary Diagnosis" directly without needing complex preprocessing.
 - **Class Balancing:** It allows us to "weight" the rare 30-day readmissions more heavily, ensuring the model doesn't simply ignore them because they only represent 11% of the data.
 - **Clinical Utility:** It generates a precise probability score (0-100%) that allows teams to triage patients into High, Medium, or Low-risk categories at discharge

As the LightGBM model is designed to learn in small increments, I used 500 estimators as a balanced starting point that is less likely to over/under fit than something like 1000 or 50. 500 estimators means that 500 different decision trees will be used. I used a low learning rate of 0.02 for better precision, higher values like 0.1 would risk underfitting by moving too quickly across the dataset.

Each decision tree looks at all columns to find the best way to split the patients to find patients that will readmit. It chooses the specific leaf (group of patients) that is currently the 'most wrong' and splits it away from its search group to reduce error (patients who have very low chance of readmitting). It repeats this process until the tree has a total of 64 leaves.

The number 64 is a deliberate choice. Num_leaves is the parameter in LightGBM which determines the complexity of the model. We used num_leaves as 64 as that is the recommended number of leaves for a model size around 100,000. Choosing a much higher number would risk making the model too specific to this particular dataset, and limit its ability to make general predictions.

2. Data Source and Data Dictionary

2.1 Dataset Overview

Source: UCI Machine Learning Repository — "Diabetes 130-US Hospitals for Years 1999-2008"

Attribute	Value
Original records	101,766 inpatient encounters
Final records	99,340 (after cleaning)
Time period	1999-2008 (10 years)
Data range	Taken from 130 US hospitals

2.2 Key Features

- Demographics: age, gender, race
- Clinical: time_in_hospital, num_lab_procedures, num_medications, number_diagnoses
- Prior history: number_inpatient, number_emergency, number_outpatient
- Diagnosis: ICD-9 codes grouped into 9 clinical categories
- Medications: 7 diabetes medications with dosage status (No/Steady/Up/Down)
- Administrative: discharge_disposition_id, payer_code, medical_specialty

2.3 Data Dictionary

Target Variable:

Variable	Type	Description
readmitted	Categorical	Original values: 'NO' (not readmitted), '>30' (readmitted after 30 days), '<30' (readmitted within 30 days)

Patient Identifiers:

Variable	Type	Description
encounter_id	Integer	Unique identifier of an encounter (hospital admission event)
patient_nbr	Integer	Unique identifier of a patient (one patient may have multiple encounters)

Demographic Features:

Variable	Type	Description
race	Categorical	Values: Caucasian, African-American, Hispanic, Asian, Other, or '?'.
gender	Categorical	Values: Male, Female, Unknown/Invalid.

age	Categorical	10-year age brackets: [0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100).
weight	Categorical	Weight in pounds (brackets). >96% missing ('?'), excluded from analysis.

Admission and Discharge Features:

Variable	Type	Description
admission_type_id	Integer	Type of admission: 1=Emergency, 2=Urgent, 3=Elective, 4=Newborn, 5=Not Available, 6=NULL, 7=Trauma Centre, 8=Not Mapped
discharge_disposition_id	Integer	Where patient was discharged: 1=Home, 2=Short-term hospital, 3=SNF, 4=ICF, 5=Other facility, 6=Home with home health, 7=Left AMA, 11=Expired, 22=Rehab, etc.
admission_source_id	Integer	Source of admission: 1=Physician Referral, 2=Clinic Referral, 3=HMO Referral, 4=Transfer from hospital, 7=Emergency Room, etc.
payer_code	Categorical	Payment source (e.g., Medicare, Medicaid, private insurance). ~40% missing.
medical_specialty	Categorical	Speciality of admitting physician (73 unique values). ~50% missing.

Clinical Encounter Features:

Variable	Type	Description
time_in_hospital	Integer	Number of days between admission and discharge (range: 1–14 days).
num_lab_procedures	Integer	Number of laboratory tests performed during the encounter.
num_procedures	Integer	Number of non-laboratory procedures (e.g., surgeries, imaging) performed.
num_medications	Integer	Number of distinct generic medications administered during encounter.
number_diagnoses	Integer	Total number of diagnoses entered for the patient.

Historical Utilisation Features ("Big 3"):

Variable	Type	Description
number_outpatient	Integer	Number of outpatient visits by the patient in the year preceding the encounter.
number_emergency	Integer	Number of emergency visits by the patient in the year preceding the encounter.
number_inpatient	Integer	Number of inpatient visits by the patient in the year preceding the encounter.

Diagnosis Features:

Variable	Type	Description
diag_1	Categorical	Primary diagnosis (ICD-9 code). Mapped to 9 clinical categories: Circulatory, Respiratory, Digestive, Diabetes, Injury, Musculoskeletal, Genitourinary, Neoplasm, Other.
diag_2	Categorical	Secondary diagnosis (ICD-9 code). Same category mapping as diag_1.
diag_3	Categorical	Additional secondary diagnosis (ICD-9 code). Same category mapping.
max_glu_serum	Categorical	Maximum glucose serum test result: None, >200, >300, Normal. Mostly missing.
A1Cresult	Categorical	HbA1c test result: None, >7, >8, Normal. Mostly missing (~83%).

Medication Features:

Variable	Type	Description
metformin, insulin, glipizide, etc.	Categorical	23 individual medication columns. Values: 'No' (not prescribed), 'Steady' (dose unchanged), 'Up' (dose increased), 'Down' (dose decreased).
change	Categorical	Whether there was any change in diabetic medications: 'No' or 'Ch' (changed).
diabetesMed	Categorical	Whether any diabetic medication was prescribed: 'Yes' or 'No'.

3. Data Cleaning

Cleaning Step	Action	Rationale
Removed the weight, max_glu_serum, and A1Cresult columns	Removed 3 columns	83-97%+ of the columns were missing, so they do not have sufficient data to be useful.
Removed the encounter_id and patient_nbr columns	Removed 2 columns	They are ID columns and not useful for predictions.
Removed Gender = Unknown/Invalid rows	Removed 3 rows	There are only a few rows like this, and they might risk being overprioritised by the model.
Removed patient encounters with Death/Hospice discharge (IDs 11,13,14,19,20,21 in the discharge_disposition_id column)	Removed 2,423 rows	Encounters who have expired cannot be readmitted, so they will unfairly affect the model.
Replace the '?' values in the medical_specialty and payer_code columns with 'Missing'	Replaced all '?' values with missing	These columns contain valuable patient information but contain significant (around 40% of the columns) missing values. Converting the '?' values allows them to still be used for analysis.
Converted age range columns to their median age	[0-10) → 5, [10-20) → 15, [20-30) → 25, [30-40) → 35, [40-50) → 45, [50-60) → 55, [60-70) → 65,	Original age is in 10-year brackets (e.g., '[70-80]') which cannot be used directly in

	[70-80) → 75, [80-90) → 85, [90-100) → 95	models; converting to midpoint creates a numeric feature that preserves ordinal relationship
Handled the '?' values in the diagnosis columns (diag_1, diag_2, diag_3) as null values	Replaced with NaN, then grouped into 'Other' category during ICD-9 mapping	Missing diagnosis codes cannot be mapped to clinical categories
Converted the '?' values in the Race column as others	Grouped into 'Other' category (3,703 encounters)	Unknown race still provides useful signal as a category
Converted emergency_admission and emergency_source columns based on the emergency admission codes	emergency_admission: 70,501 (71.0%) emergency vs 28,839 (29.0%) non-emergency	Simplifies admission type into clinically meaningful binary flags
Converted diagnosis columns to categorical variables based on their ICD-9 codes.	9 categories: Circulatory, Respiratory, Diabetes, Digestive, Injury, Musculoskeletal, Genitourinary, Neoplasms, Other	Raw ICD-9 codes are too granular; clinical groupings are more meaningful
Created binary targets for the target variable (readmissions < 30 and > 30 as 1 and No as 0 for the first model, and < 30 as 1 and > 30 and No as 0 for the second model).	Model 1: 47.1% positive (46,816). Model 2: 11.4% positive (11,314) — requires class balancing	Two prediction tasks: any readmission vs. early (30-day) readmission
Filtered medication columns to keep only those with >5% prescription rate, retained as categorical variables (No/Steady/Up/Down)	Retained 7 medications: insulin (49.2%), metformin (20.5%), glipizide (15.0%), glyburide (14.0%), pioglitazone (8.0%), rosiglitazone (6.9%), glimepiride (6.5%).	23 original medication columns, but most have <5% prescription rate (too sparse to be predictive). Keeping as categories

	Dropped 16 sparse medications.	preserves dosage change information.
--	--------------------------------	--------------------------------------

4. Feature Engineering

4.1 Target Variables

Any Readmission (Primary Model)

- Not Readmitted (0): 52,524 patients (52.9%)
- Readmitted (1): 46,816 patients (47.1%)

Near-balanced distribution — no class balancing required.

Early Readmission <30 Days (Secondary Model)

- Not Early Readmitted (0): 88,026 patients (88.6%)
- Early Readmitted (1): 11,314 patients (11.4%)

Severe class imbalance (7.8:1 ratio) — requires class balancing.

Categorical variables:

- Converted the medical_specialty, payer_code, race, gender to categorical columns which can be used by the LightGBM model.
- For the logistic regression and random forest model, I converted the categorical columns into dummy columns.

4.2 Final Feature Set

Total features: 28 (for LightGBM with native categoricals)

- Numeric: 10 features (age, time_in_hospital, lab procedures, etc.)
- Categorical: 14 features (race, gender, medications, diagnosis groups, etc.)
- Binary: 4 features (emergency_admission, change, diabetesMed, etc.)

5. Exploratory Data Analysis

5.1 Target Distribution

Any readmission: Near-balanced (47% vs 53%)

30-day readmission: Severely imbalanced (11% vs 89%)

5.2 Readmission by Discharge Disposition

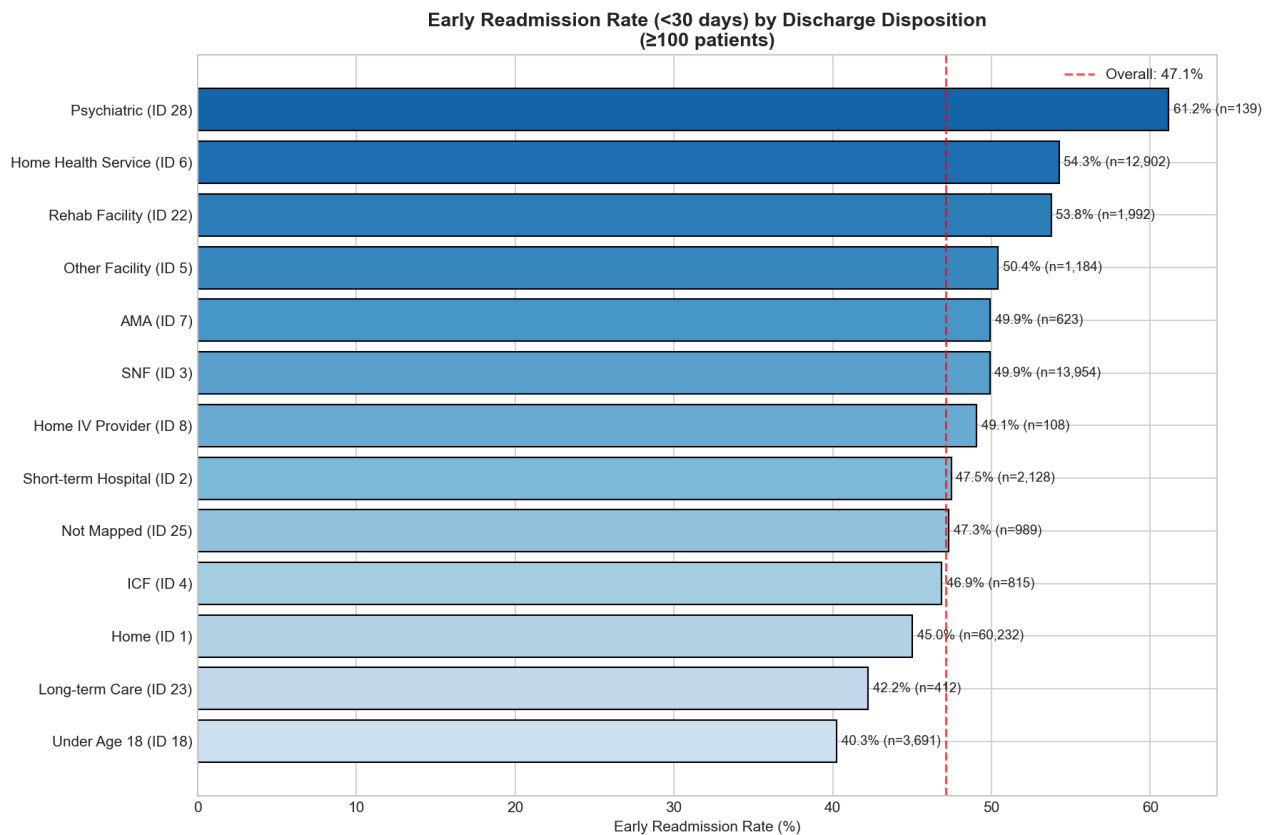


Figure 1: Readmission rate by discharge disposition

Psychiatric patients are high-risk though they occupy a small portion of the dataset. Home Health Service patients are high-risk with a significant proportion of the dataset. This could be because they are without care support at home, and will thus likely be readmitted when their condition worsens. Conversely, those sent to Short-term Hospitals or Rehab Facilities show lower readmission rates, likely due to continued clinical oversight.

5.3 Readmission by Primary Diagnosis Group

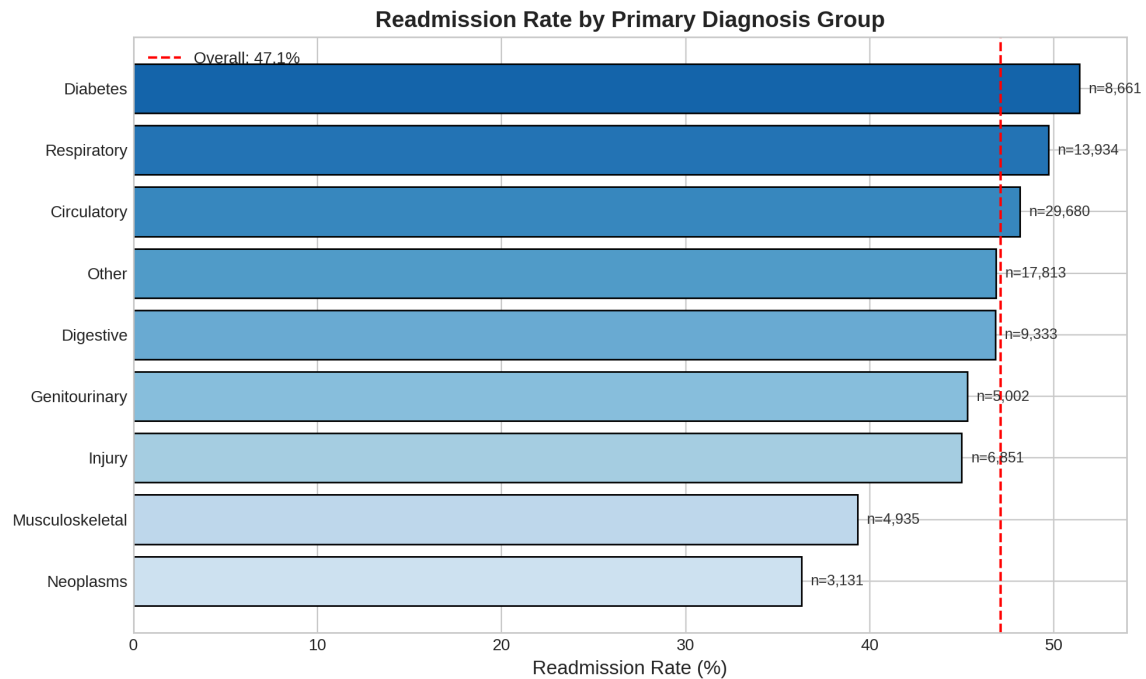


Figure 2: Readmission rate by primary diagnosis (ICD-9 grouped)

Readmission rates vary slightly by diagnosis. Diabetes as a primary diagnosis carries a 52.6% readmission rate, the highest out of all diagnosis. Circulatory conditions, the most common primary diagnosis, sit near the overall average. Injury and Musculoskeletal conditions have the lowest rates. As acute episodes with defined recovery paths, they present fewer chronic management challenges than other conditions.

5.4 Readmission by Medication Variables

Readmission Rates by Medication Variables

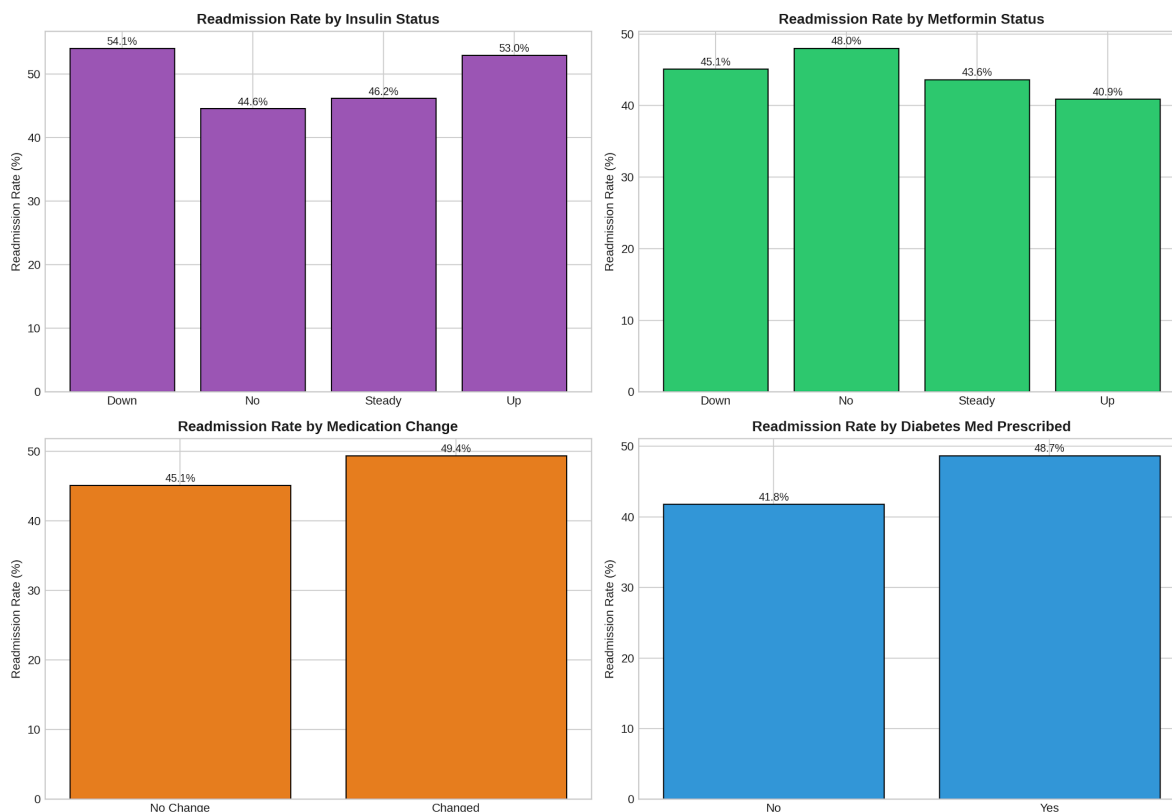


Figure 3: Readmission rates by insulin, metformin, medication change, and diabetes medication status

Medication data reveals that dose instability is a stronger predictor than medication type alone. Patients whose insulin dose was increased ("Up") have a 53.0% readmission rate, compared to 43.6% for those not on insulin. This pattern holds for metformin. Any dosage adjustment during an encounter correlates with higher readmission (49.4% vs 45.1%). Furthermore, patients prescribed diabetes medication have a higher readmission rate (48.7%) than those who were not (41.8%). The fact that all increases in medication results in an increase in readmission rate aligns with the earlier finding that more treatment and clinical procedures depicts a patient with a more serious condition, and thus a patient who is more likely to readmit.

5.5 LightGBM Feature Importance (Top 20)

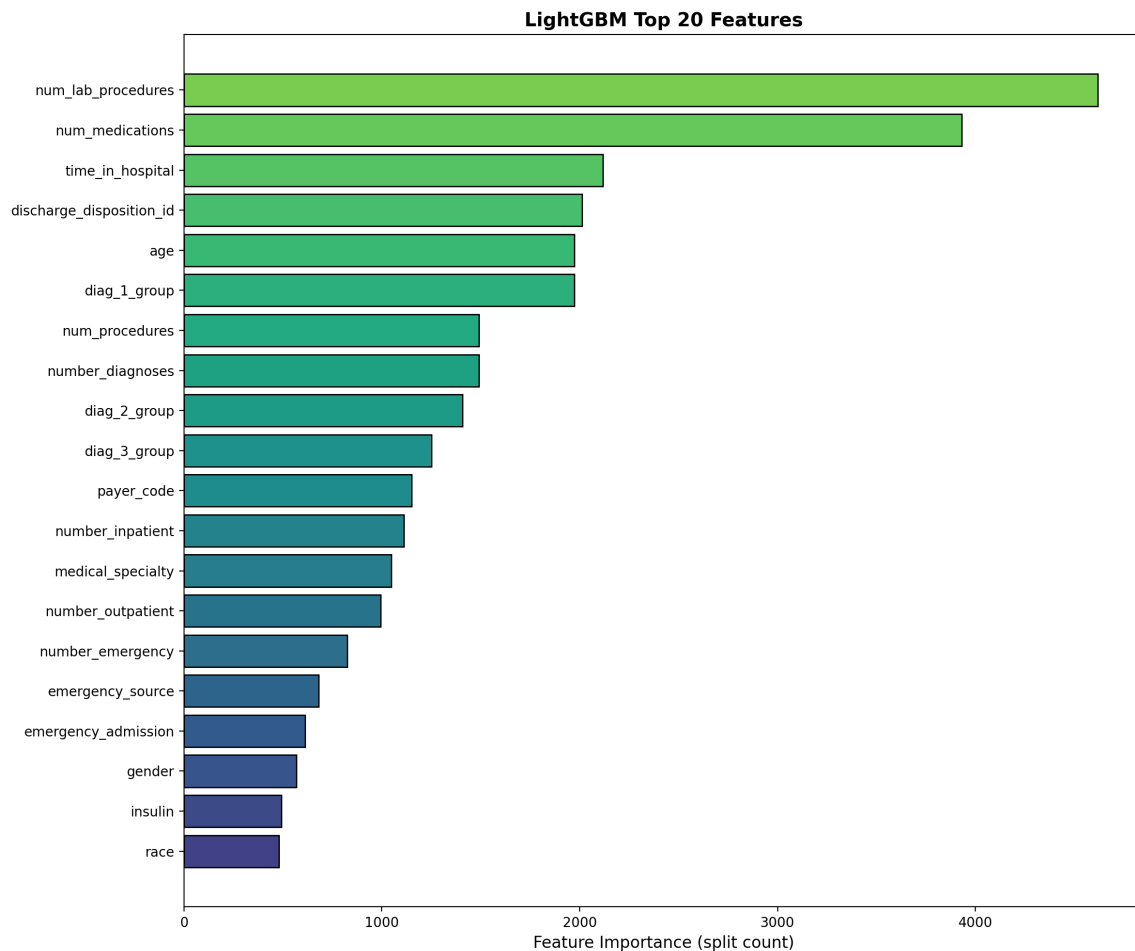


Figure 4: LightGBM top 20 features ranked by split count

The LightGBM feature importance is calculated by counting how many times a specific feature was used to make a split across all 500 trees. The top factors were used to split and reduce prediction error the most, and that is why they are considered the most valuable for predicting if a patient will readmit or not.

The plot identifies `num_lab_procedures` as the most influential variable, followed by `num_medications` and `discharge_disposition_id`. Clinical encounter features, such as lab counts and time in hospital, dominate the model, indicating that the intensity of the current stay is a primary predictor. Notably, `number_inpatient` history also ranks in the top 10. Because LightGBM handles categorical data natively, diagnosis groups appear as single, efficient entries rather than numerous dummy variables.

6. Modelling Approach

6.1 Train-Test Split

80% training (79,472) / 20% test (19,868), stratified by target.

6.2 Algorithms

- Logistic Regression: Linear model, 88 one-hot features, StandardScaler
- Random Forest: 100 trees, max_depth=10, 88 one-hot features
- LightGBM: 500 trees, num_leaves=64, learning_rate=0.02, native categoricals (28 features)

6.3 Class Balancing for 30-Day Model

The 30-day readmission target has severe class imbalance (7.8:1). Strategy:

3:1 Moderate Class Weighting

To provide some balance to the dataset, I gave a weightage of 3 to the not-early readmitted within 30 days values. This weighting is preferable to a more aggressive ratio like 8, which could unrealistically overskew results and generate excessive false positives. Since 30-day readmissions remain relatively rare occurrences (11.4%), this 3:1 ratio provides a more realistic and balanced dataset for the model.

Class	Count	Weight
Not Early Readmitted (0)	88,026	1.0
Early Readmitted <30 days (1)	11,314	3.0

Effective ratio after weighting: 2.6:1 (reduced from 7.8:1)

Threshold Tuning

I used the threshold number as 0.3 rather than the baseline of 0.5. This means that the model will flag patients at a 0.3 as likely to readmit rather than 0.5. This means that the model is more likely to identify at-risk individuals, providing a generous estimate of potential readmissions. This approach prioritises clinical safety; because the resource cost of missing a real admission is high, we aim to minimise false negatives.

Optimal threshold: 0.3 (vs. default 0.50)

Metric	Baseline Model	LightGBM Tuned (0.3)
Accuracy	0.8861	0.7084
Precision	0.0000	0.1946
Recall	0.0000	0.4971
F1-Score	0.5000	0.2792

Tuning increases recall from 11% to 49.7% — catching 4x more actual early readmissions.

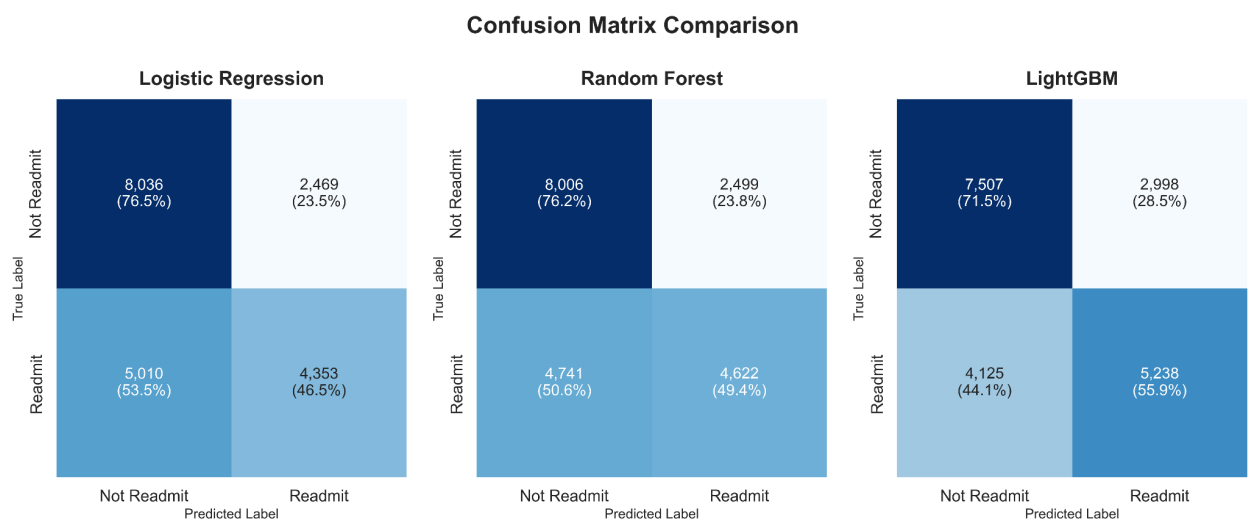
7. Results: Model 1: Any Readmission

7.1 Algorithm Comparison

Metric	LogReg	Random Forest	LightGBM	Winner
Accuracy	0.624	0.636	0.642	LightGBM
Precision	0.638	0.649	0.636	RF
Recall	0.465	0.494	0.559	LightGBM
F1-Score	0.538	0.561	0.595	LightGBM
AUC-ROC	0.669	0.685	0.692	LightGBM

Selected: LightGBM — best on AUC, F1, and Recall.

7.2 Confusion Matrix Comparison (Model 1)



LightGBM got the best score for catching actual readmissions by about 6%, which is a significant number in a critical clinical context. While it had a higher rate of false positives (around 5%), the cost of missing out actual readmissions is much higher than slightly overpredicting readmissions. These false positives also offer us an opportunity for clinical exploration (studying the patients who were likely to be readmitted and did not to understand).

7.3 Score Analysis

While the score is not as high, the model does also assign each encounter with a risk score. This means that even if a patient is assigned a score of 0.45, they will be flagged as unlikely to readmit. In actual implementation, the patient with a score of 0.45 will still be flagged to clinical teams to offer extra support to.

Additionally, the scores are a reflection of the complexity of healthcare, and that a patient's care journey is determined by several physical or socio-economic factors that are not captured on their data sheet.

8. Results: Model 2: 30-Day Readmission

The 30-day model uses 3:1 class weighting and threshold tuning to address severe class imbalance.

8.1 Algorithm Comparison (with Class Weighting)

Metric	LogReg	Random Forest	LightGBM*	Winner
Accuracy	0.873	0.886	0.708	RF
Precision	0.324	0.475	0.194	RF
Recall	0.105	0.013	0.497	LightGBM
F1-Score	0.159	0.025	0.280	LightGBM
AUC-ROC	0.651	0.666	0.671	LightGBM

**LightGBM uses tuned threshold (0.3); others use default (0.50)*

LightGBM with threshold tuning catches almost 50% of early readmissions vs. only 1-10% for other models.

8.2 Confusion Matrix Comparison

Percentage of actual early readmissions correctly identified (True Positive Rate):

Model	Recall (TPR)
Logistic Regression	10.5%
Random Forest	1.3%
LightGBM (threshold=0.30)	49.7%

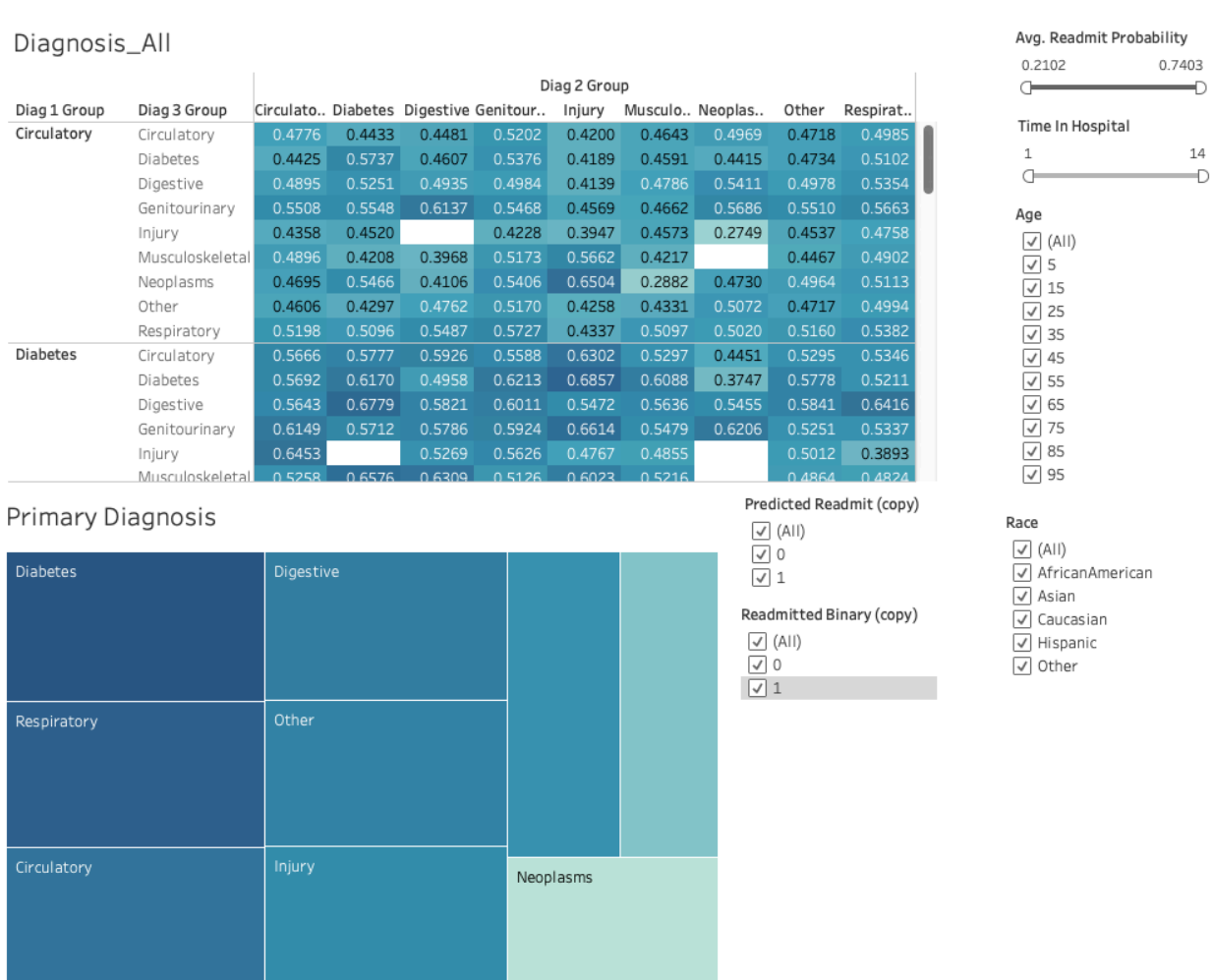
8.3 Trade-off Analysis

The tuned threshold sacrifices accuracy (88% → 70%) and precision (36% → 19%) to achieve much higher recall (11% → 45%). This trade-off is appropriate because:

- Clinical priority: Missing an early readmission is more costly than being overly prepared.
- Follow-up costs and support actions would generally to be cheaper than a readmission.
- Patient impact: Early readmissions often indicate care gaps that need urgent attention.

9. Interactive Dashboard Application

An interactive Tableau dashboard allows clinical and hospital planning teams to explore readmission patterns across the patient population. The tool converts model outputs into actionable insights through two primary views:



9.1 Key Visualisation Components

- **Diagnosis Heatmap:** This view displays the average predicted readmission probability for combinations of primary, secondary, and tertiary diagnoses.
- **Risk Identification:** Cells are colour-coded to highlight high-risk combinations. For instance, patients with a primary diagnosis of Diabetes and a secondary Genitourinary condition show readmission probabilities exceeding 60%.
- **Low-Risk Profiles:** Conversely, patients with Injury and Musculoskeletal combinations often fall below 40%.

- **Primary Diagnosis Treemap:** This provides a proportional view of risk level, with size and colour representing higher risk.

9.2 Targeted Analysis and Filtering

Interactive filters allow users to slice data by specific clinical and demographic dimensions:

- **Risk Stratification:** Filter by readmission probability range (0-100%) or predicted status to immediately find groups of high/ low risk.
- **Clinical Factors:** Segment by time in hospital or age group.
- **Demographics:** We can use the filters to check on different age, race or gender groups to find abnormalities or groups with particularly high/ low risk.
- **Outcome Validation:** Compare predicted scores against actual readmission outcomes.

For example, a care coordinator can filter for patients aged 75+ with predicted risk scores above 0.70 to identify the highest-risk elderly cohort for intensive 48-hour follow-up calls.

10. Conclusions and Recommendations

10.1 Summary

Two models were developed for different clinical use cases:

- Any readmission model: AUC 0.692, catches 55.9% of readmissions — good for general risk stratification.
- 30-day model: AUC 0.671, catches 49.7% with tuned threshold — gives hospital planners a forecast to plan for beds and resource allocation.

10.2 Recommendations

1. Deploy both models: Use any-readmission for general triage, 30-day for urgent cases
2. Allow threshold tuning: Let clinical teams adjust based on their capacity
3. Validate on Singapore data: Retrain with local features (polyclinic visits, subsidies)
4. Build feedback loop: Track outcomes to improve models over time

10.3 Future Models

For Singapore's focus on preventive healthcare, a model which can predict ADMISSIONS from a dataset of diabetes patients would be extremely beneficial both for patients, care workers and hospital management. With around roughly 1 in 5 hospital admissions in Singapore being diabetes related, being able to score the risk for diabetic patients through a machine learning model would be extremely valuable for preventive care.

Additional data points, mainly weight, lab results and even potentially socio-economic conditions (EG: if the patient is living alone) would allow for greater accuracy in such a model.

11. Limitations

11.1 Data Limitations

- US data from 1999-2008 will not translate directly to Singapore 2026.
- No weight data.
- No social determinants (housing, family support, financial status).
- No lab values (HbA1c, glucose levels).

11.2 Model Limitations

- Any-readmission AUC 0.69 is acceptable but not excellent. It should be used only as a rough guideline and forecast.
- 30-day model precision is low (19%) due to class imbalance.
- Correlation should not be taken as causation.

11.3 Implementation Considerations

- Workflow integration: Must fit into the discharge process
- Alert fatigue: Too many flags may be ignored
- Equity: Monitor for differential performance across groups
- Maintenance: Periodic retraining required

— *End of Technical Report* —

Citations

Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Beis, D. (2014). *Diabetes 130-US hospitals for years 1999-2008* [Data set]. UCI Machine Learning Repository.

Ministry of Health & Health Promotion Board. (2025, October 17). *National Population Health Survey (NPHS) 2024 report*. Ministry of Health Singapore.

Ministry of Health Singapore. (n.d.). *Beds occupancy rate (BOR)*. Retrieved February 24, 2026, from [https://www.moh.gov.sg/others/resources-and-statistics/healthcare-institution-statistics-beds-occupancy-rate-\(bor/\)](https://www.moh.gov.sg/others/resources-and-statistics/healthcare-institution-statistics-beds-occupancy-rate-(bor/))